

Modbusprotocol communication protocol

The register address and corresponding function code table are as follows:

Register address	Name	Function	Function code 0X03	Function code 0X06	Function code 0X0A	Function code 0X0B	Function code 0X0C	Assignment range
0X0000	Remote Mode	Remote operation mode 0: Local operation 1: Remote operation						0X0000~0X0001
0X0001	CH1 Vmax	CH1 Maximum voltage setting 0~31V / 0~61V						0X0000~0X7918 / 0X0000~0XEE48
0X0002	CH2 Vmax	CH2 Maximum voltage setting 0~31V / 0~61V						0X0000~0X7918 / 0X0000~0XEE48
0X0003	CH3 Vmax	CH3 Maximum voltage setting 0~6V						0X0000~0X1770
0X0004	CH1 Amax	CH1 Maximum current setting 0~3.1A / 0~6.1A						0X0000~0X0C1C / 0X0000~0X17D4
0X0005	CH2 Amax	CH1 Maximum current setting 0~3.1A / 0~6.1A						0X0000~0X0C1C / 0X0000~0X17D4
0X0006	CH3 Amax	CH3 Maximum current setting 0~3.1A						0X0000~0X0C1C
0X0007	CH1 Vset	CH1 setting voltage 0~31V / 0~61V						0X0000~0X7918 / 0X0000~0XEE48
0X0008	CH1 Aset	CH1 setting current 0~3.1A / 0~6.1A						0X0000~0X0C1C / 0X0000~0X17D4
0X0009	CH2 Vset	CH2 setting voltage 0~31V / 0~61V						0X0000~0X7918 / 0X0000~0XEE48
0X000A	CH2 Aset	CH2 setting current 0~3.1A / 0~6.1A						0X0000~0X0C1C / 0X0000~0X17D4

0X000B	CH3 Vset	CH3 setting voltage 0~6V						0X0000~0X1770
0X000C	CH3 Aset	CH3setting current 0~3.1A						0X0000~0X0C1C
0X000D	CH1 OVPset	CH1 overvoltage setting 0~32V / 0~62V						0X0000~0X7D00 / 0X0000~0XF230
0X000E	CH2 OVPset	CH2 overvoltage setting 0~32V/0~62V						0X0000~0X7D00 / 0X0000~0XF230
0X000F	CH3 OVPset	CH3 over voltage setting 0~7V						0X0000~0X1B58
0X0010	CH1 OutTimer	CH1 timing output time						0X0000~0X8CA0
0X0011	CH2 OutTimer	CH2 timing output time						0X0000~0X8CA0
0X0012	CH2 OutTimer	CH3 timing output time						0X0000~0X8CA0
0X0013	CH1 TIM On/OFF	CH1 Timing output state setting 0: close 1:open						0X0000~0X0001
0X0014	CH2 TIM On/OFF	CH2Timing output state setting 0: close 1:open						0X0000~0X0001
0X0015	CH3 TIM On/OFF	CH3Timing output state setting 0: close 1:open						0X0000~0X0001
0X0016	OutState	Power-on output status 0: output close 1:Maintain the state before shutdown						0X0000~0X0001
0X0017	OutParamState	Boot parameter status 0:Factory default 1:Pre-shutdown parameters are maintained						0X0000~0X0001
0X0018	KnobState	Code switch status 0:unlock 1:lock						0X0000~0X0001
0X0019	BuzzerState	Buzzer status 0:close 1:open						0X0000~0X0001
0X001A	MemoryGroup	storage group						0X0001-0X0004
0X001B	LockState	keyboard lock status 0:unlock 1:lock						0X0000-0X0001
0X001C	CombModeState	Combined state 0:independent 1:serial 2:parallel 3:tracking						0X0000-0X0003

0X001D	CombChunnelState	Combined channel 0:CH1+CH2 1: CH2+CH32:CH1+CH2+CH3						0X0000-0X0002
0X001E	CH1 On/Off	CH1Channel output state setting 0:close 1:output						0X0000~0X0001
0X001F	CH2 On/Off	CH2Channel output state setting 0:close 1:output						0X0000~0X0001
0X0020	CH3 On/Off	CH3Channel output state setting 0:close 1:output						0X0000~0X0001
0X0021	ALL On/Off	Power total output state setting 0:close 1:output						0X0000~0X0001
0X0022	CH1 Vout	CH1 actual voltage 0~31V / 0~61V						0X0000~0X7918 / 0X0000~0XEE48
0X0023	CH1 Aout	CH1 actual current 0~3.1A / 0~6.1A						0X0000~0X0C1C / 0X0000~0X17D4
0X0024	CH1 CV/CC	CH1 Constant voltage and constant current status flag 0:CV 1:CC						0X0000~0X0001
0X0025	CH2 Vout	CH2actual voltage 0~31V / 0~61V						0X0000~0X7918 / 0X0000~0XEE48
0X0026	CH2 Aout	CH2actual current 0~3.1A / 0~6.1A						0X0000~0X0C1C / 0X0000~0X17D4
0X0027	CH2 CV/CC	CH2Constant voltage and constant current status flag 0:CV 1:CC						0X0000~0X0001
0X0028	CH3 Vout	CH3 actual voltage 0~6V						0X0000~0X1770
0X0029	CH3 Aout	CH3 actual current 0~3.1A						0X0000~0X0C1C
0X002A	CH3 CV/CC	CH3Constant voltage and constant current status flag 0:CV 1:CC						0X0000~0X0001
0X002B	CAL Year	Calibration Date: Calibration Year						0X0000~0X0802
0X002C	CAL Month/Day	Calibration date: high byte is month, low byte is day						0X01~0X0C,0X01~0X1F

0X002D	Factory	Manufacturer: A total of 10 bytes of ASCII code						0X20~0X7E,0X20~0X7E
0X0031								0X20~0X7E,0X20~0X7E
0X0032	Model	Product model: A total of 12 bytes ASCII code						0X20~0X7E,0X20~0X7E
0X0037								0X20~0X7E,0X20~0X7E
0X0038	SN	Product serial number: a total of 14 bytes of ASCII code						0X20~0X7E,0X20~0X7E
0X003E								0X20~0X7E,0X20~0X7E
0X003F	Hardware Version	Hardware version:A total of 2 bytes ASCII code						0X20~0X7E,0X20~0X7E
0X0040	Software Version	Software version: A total of 2 bytes ASCII code						0X20~0X7E,0X20~0X7E
0X005B	USER ADDR 1	User storage point 1 0:recall 1:save						0X0000~0X0001
0X0064	USER ADDR 10	User storage point 10 0:recall 1:save						0X0000~0X0001
0X0065	CH1 V ADJ 1	CH1 Voltage calibration point 1						0X0000~0XFFFF
0X006E	CH1 V ADJ 10	CH1Voltage calibration point 10						0X0000~0XFFFF
0X00C9	CH2V ADJ 1	CH2Voltage calibration point 1						0X0000~0XFFFF
0X00D2	CH2V ADJ 10	CH2Voltage calibration point 10						0X0000~0XFFFF
0X012D	CH3V ADJ 1	CH3Voltage calibration point 1						0X0000~0XFFFF
0X0136	CH3V ADJ 10	CH3Voltage calibration point 10						0X0000~0XFFFF
0X0191	CH1 A ADJ 1	CH1current calibration point 1						0X0000~0XFFFF
0X019A	CH1 A ADJ 10	CH1current calibration point 10						0X0000~0XFFFF
0X01F5	CH2A ADJ 1	CH2current calibration point 1						0X0000~0XFFFF
0X01FE	CH1 A ADJ 10	CH2current calibration point 10						0X0000~0XFFFF
0X0259	CH3A ADJ 1	CH3current calibration point 1						0X0000~0XFFFF
0X0263	CH3A ADJ 10	CH3current calibration point 10						0X0000~0XFFFF

Note: All operations take effect in remote operation mode (register 0X0000 value is 0X0001)!

*Based on local operating mode after power cycle!

The function code table and interval time are as follows:

Function code	Corresponding function	Time between two operations
0X03	Read one or more registers consecutively	N * 5ms
0X06	Write a single register command	(register 0x0000~0x0021)10ms, (register 0x0065~0x02BC)300ms
0X0A	Write one or more register commands in succession	N * 5ms
0X0B	A single configuration corresponds to a calibration point and returns the calibration point value	10ms
0X0C	Save one or more calibration points in a row	N * 10ms

Note: N is the number of registers

The communication protocol format is as follows:

Function code 0X03

PC send: 8 Byte

Address	Function code	Start address high byte	Start address low byte	End address high byte	End address low byte	CRC16 Check code high byte	CRC16 Check code low byte
0XFE	0X03						

power return: 8+N*2 Byte

Address	Function code	Start address high byte	Start address low byte	End address high byte	End address low byte	Return data high byte	Return data low byte	Return data N+1 high byte	Return data N+1 low byte	CRC16Check code high byte	CRC16Check code low byte
0XFE	0X03										

*address range:(0X0000~0X0040)

Example 1: Read a single register status(0X0000)

PC send: FE 03 00 00 00 00 C5 51 (start address is 0X0000, end address is 0X0000, CRC16The verification result is 0X0551)

Power return: FE 03 00 00 00 00 01 03 3C (0X0001Indicates remote operation mode, 0X033C is CRC16 calibration result)

Example 2: Read Multiple Register Status(0X0001~0X0003)

PC send: FE 03 00 01 00 03 04 40 (start address is 0X0001, ending address is 0X0003, CRC16 checking result is 0X0440)

Power return: FE 03 00 01 00 03 03E8 07D0 0B B8 EB 98 (0X03E8 is register 0X0001 value, 0X07D0 is register 0X0002 value, 0X0BB8is register 0X0003 value, 0XEB98 is CRC16 calibration result)

Function code 0X06

PC send: 8 Byte

Address	Function code	Write address high byte	Write address low byte	Write data high byte	Write data low byte	CRC16Check code high byte	CRC16Check code low byte
0XFE	0X06						

Power return: 8Byte

Address	Function code	Write address high byte	Write address low byte	Write data high byte	Write data low byte	CRC16Check code high byte	CRC16Check code low byte
0XFE	0X06						

*address range: (0X0000~0X0021) (0X002B~0X02BC)

Example 1: Read a single register status (0X0000)

PC send: FE 06 00 00 00 01 05 5C (The write address is 0X0000, the write data is 0X0001, and the CRC16 check result is 0X055C)

Power return: FE 06 00 00 00 01 05 5C (The write address is 0X0000, the write data is 0X0001, and the CRC16 check result is 0X055C)

Function code 0X0A

PC send: 8+N*2 Byte

Address	Function code	Starting address high byte	Starting address low byte	End address high byte	End address low byte	Write data high byte	Write data low byte	Write data N+1 high byte	Write data N+1 low byte	CRC16 check code high byte	CRC16 check code low byte
0XFE	0X0A										

Power return: 8+N*2 Byte

Address	Function code	Starting address high byte	Starting address low byte	End address high byte	End address low byte	Write data high byte	Write data low byte	Write data N+1 high byte	Write data N+1 low byte	CRC16 check code high byte	CRC16 check code low byte
0XFE	0X0A										

*address range: (0X0000~0X0021) (0X002B~0X0040)

Example 1: Write single register status (0x0001)

PC send: FE 0A 00 01 00 0103 E8 7D 08 (Start address is 0x0001, end address is 0x0001, write data is 0x03E8, CRC16 check result is 0x7D08)

Power return: FE 0A 00 01 00 01 03 E8 7D 08 (Start address is 0x0001, end address is 0x0001, write data is 0x03E8, CRC16 check result is 0x7D08)

Example 2: Write multiple registers status (0x0001~0x0003)

PC send: FE 0A 00 01 00 03 03 E8 07 D0 0B B8 EB 98 (Start address is 0x0001, end address is 0x0003, CRC16 check result is 0xEB98)

Power return: FE 0A 00 01 00 03 03 E8 07 D0 0B B8 EB 98 (0x03E8 is the value of register 0x0001, 0x07D0 is the value of register 0x0002, 0x0BB8 is the value of register 0x0003)

Function code 0X0B

PC send: 8 Byte

Address	Function code	Configuration address high byte	Configuration address low byte	Write password high byte	Write password low byte	CRC16 check code high byte	CRC16 check code low byte
0XFE	0X0B			0X88	0X66		

Power return: 8Byte

Address	Function code	Configuration address high byte	Configuration address low byte	Return data high byte	Return data low byte	CRC16 check code high byte	CRC16 check code low byte
0XFE	0X0B						

*address range: 0X0065~0X02BC The returned data is the calibration point parameter value.

Example 1: Configure the first calibration point (0x0065) of CH1 voltage

PC sends: FE 0B 00 65 88 66 31 46 (Configure calibration point address as 0x0065, write password as 0x8866, CRC16 check result as 0x3146)

Power supply returns: FE 0B 00 65 00 0A 1C 20 (Configure calibration point address as 0x0065, calibration voltage as 0x000A, CRC16 check result as 0x1C20)

Function code 0X0C

PC send: 10 Byte

Address	Function code	Start address high byte	Start address low byte	End address high byte	End address low byte	Write password high byte	Write password low byte	CRC16 check code high byte	CRC16 check code low byte
0XFE	0X0C					0X88	0X66		

PC send: 10 Byte

Address	Function code	Start address high byte	Start address low byte	End address high byte	End address low byte	Return data high byte	Return data low byte	CRC16 check code high byte	CRC16 check code low byte
0XFE	0X0C								

*address range: 0X0065~0X02BC Return status 0x0001: Save successful; Return status 0x0000: Save failed.

Example 1: Save the first calibration point (0x0065) of CH1 voltage

PC sends: FE 0B 00 65 88 66 3E CE (Save calibration point start address 0x0065, end address 0x0065, write password 0x8866, CRC16 check result 0x76BC)

Power supply returns: FE 0B 00 65 00 01 D4 E9 (Save calibration point start address 0x0065, end address 0x0065, 0x0001 indicates save successful, CRC16 check result 0xD4E9)

Example 2: Save all calibration points (0x0065~0x00C8) of CH1 voltage

PC sends: FE 0B 00 65 00 C8 88 66 DF 5F (Save calibration point start address 0x0065, end address 0x00C8, write password 0x8866, CRC16 check result 0xDF5F)

Power supply returns: FE 0B 00 65 00 C8 00 01 35 78 (Save calibration point start address 0x0065, end address 0x00C8, 0x0001 indicates save successful, CRC16 check result 0x3578)

[illegible]

```
{
0x00, 0xC0, 0xC1, 0x01, 0xC3, 0x03, 0x02, 0xC2, 0xC6, 0x06, 0x07, 0xC7,
0x05, 0xC5, 0xC4, 0x04, 0xCC, 0x0C, 0x0D, 0xCD, 0x0F, 0xCF, 0xCE, 0x0E,
```

```
0x0A, 0xCA, 0xCB, 0x0B, 0xC9, 0x09, 0x08, 0xC8, 0xD8, 0x18, 0x19, 0xD9,
0x1B, 0xDB, 0xDA, 0x1A, 0x1E, 0xDE, 0xDF, 0x1F, 0xDD, 0x1D, 0x1C, 0xDC,
0x14, 0xD4, 0xD5, 0x15, 0xD7, 0x17, 0x16, 0xD6, 0xD2, 0x12, 0x13, 0xD3,
0x11, 0xD1, 0xD0, 0x10, 0xF0, 0x30, 0x31, 0xF1, 0x33, 0xF3, 0xF2, 0x32,
0x36, 0xF6, 0xF7, 0x37, 0xF5, 0x35, 0x34, 0xF4, 0x3C, 0xFC, 0xFD, 0x3D,
0xFF, 0x3F, 0x3E, 0xFE, 0xFA, 0x3A, 0x3B, 0xFB, 0x39, 0xF9, 0xF8, 0x38,
0x28, 0xE8, 0xE9, 0x29, 0xEB, 0x2B, 0x2A, 0xEA, 0xEE, 0x2E, 0x2F, 0xEF,
0x2D, 0xED, 0xEC, 0x2C, 0xE4, 0x24, 0x25, 0xE5, 0x27, 0xE7, 0xE6, 0x26,
0x22, 0xE2, 0xE3, 0x23, 0xE1, 0x21, 0x20, 0xE0, 0xA0, 0x60, 0x61, 0xA1,
0x63, 0xA3, 0xA2, 0x62, 0x66, 0xA6, 0xA7, 0x67, 0xA5, 0x65, 0x64, 0xA4,
0x6C, 0xAC, 0xAD, 0x6D, 0xAF, 0x6F, 0x6E, 0xAE, 0xAA, 0x6A, 0x6B, 0xAB,
0x69, 0xA9, 0xA8, 0x68, 0x78, 0xB8, 0xB9, 0x79, 0xBB, 0x7B, 0x7A, 0xBA,
0xBE, 0x7E, 0x7F, 0xBF, 0x7D, 0xBD, 0xBC, 0x7C, 0xB4, 0x74, 0x75, 0xB5,
0x77, 0xB7, 0xB6, 0x76, 0x72, 0xB2, 0xB3, 0x73, 0xB1, 0x71, 0x70, 0xB0,
0x50, 0x90, 0x91, 0x51, 0x93, 0x53, 0x52, 0x92, 0x96, 0x56, 0x57, 0x97,
0x55, 0x95, 0x94, 0x54, 0x9C, 0x5C, 0x5D, 0x9D, 0x5F, 0x9F, 0x9E, 0x5E,
0x5A, 0x9A, 0x9B, 0x5B, 0x99, 0x59, 0x58, 0x98, 0x88, 0x48, 0x49, 0x89,
0x4B, 0x8B, 0x8A, 0x4A, 0x4E, 0x8E, 0x8F, 0x4F, 0x8D, 0x4D, 0x4C, 0x8C,
0x44, 0x84, 0x85, 0x45, 0x87, 0x47, 0x46, 0x86, 0x82, 0x42, 0x43, 0x83,
0x41, 0x81, 0x80, 0x40
};
```

```
unsigned int crc16(unsigned char *DData,unsigned char len)
{
    unsigned char CRCHi = 0xFF;
    unsigned char CRCLo = 0xFF;
    unsigned int wIndex;
```

```
unsigned int CRC_DData;  
while(len--)  
{  
    wIndex = CRCLo ^ *DData++;  
    CRCLo = CRCHi ^ CRCHTalbe[wIndex];  
    CRCHi = CRCLTalbe[wIndex];  
}  
CRC_DData = CRCHi;  
CRC_DData<<= 8;  
CRC_DData |=CRCLo;  
    return CRC_DData;  
}
```



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